



Chromatin Binding Regulates Phase Behavior and Morphology of Condensates Formed by Prion-like Domains [☆]

Anushka Supakar^{1,†}, Richoo B. Davis^{2,†}, Subhadip Biswas^{3,†}, Sean Yang³, Davit A. Potoyan^{3,*}, and Priya R. Banerjee^{1,2,*}

1 - Department of Biological Sciences, The State University of New York at Buffalo, Buffalo, NY 14260, USA

2 - Department of Physics, The State University of New York at Buffalo, Buffalo, NY 14260, USA

3 - Department of Chemistry, Iowa State University, Ames, IA 50011, USA

Correspondence to Davit A. Potoyan and Priya R. Banerjee: Department of Physics, The State University of New York at Buffalo, Buffalo, NY 14260, USA. (P.R. Banerjee); Davit A. Potoyan, Department of Chemistry, Iowa State University, Ames, IA 50011, USA (D.A. Potoyan). potoyan@iastate.edu (D.A. Potoyan), prbanerj@buffalo.edu (P.R. Banerjee)

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Abstract

Many transcription factors (TFs) contain intrinsically disordered regions (IDRs) and are thought to form biomolecular condensates in the cell nucleus. These proteins can be conceptualized as block copolymers, with the IDRs driving both homotypic and heterotypic protein–protein interactions and the DNA-binding domain (DBD) mediating heterotypic interactions with chromatin. While *in vitro* studies have predominantly reported micron-scale, spherical condensates in the absence of chromatin, TF condensates in live cells exhibit strikingly different behavior, adopting diverse, nanoscale, often aspherical morphologies and displaying sub-diffusive dynamics. Here, using engineered fusion proteins with tunable IDR-DBD architectures, we show that this distinct phase behavior can arise from TF-chromatin interactions. Specifically, we fused the prion-like domain (PLD) of the SS18 subunit from the mammalian SWI/SNF complex, a domain known to drive homotypic phase separation, to the DBD of the pioneer factor FOXA1. While SS18^{PLD} on overexpression forms large, spherical condensates in cells, its fusion with FOXA1^{DBD} leads to condensates that re-localize to chromatin, adopt aspherical morphologies, and exhibit chromatin-wetting behavior. Disruption of DBD-chromatin binding shifts condensate morphology toward a mixed or spherical state, implicating chromatin affinity as a key regulator of condensate coarsening and spatial organization. Coarse-grained simulations recapitulate these observations, revealing a finely balanced interplay between PLD-PLD and DBD-DNA interactions that collectively determine condensate dynamics and structure. Together, our findings demonstrate that chromatin binding is a critical modulator of transcriptional condensate behavior *in vivo*.

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Introduction

The nucleus serves as the central hub for gene expression, orchestrating the activity of various transcriptional proteins to facilitate RNA production. It contains spatially separated

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sub-compartments that carry out distinct functions, which are thought to form through biomolecular condensation [1–4]. In the past decade, phase-separation of proteins and nucleic acids has emerged as the underlying thermodynamic driving process for forming biomolecular condensates such as the transcriptional hubs at super-enhancer sites and the nucleolus among others [5–7]. Recent advancements highlight the importance of multivalent interactions, sometimes facilitated by intrinsically disordered regions (IDRs) within transcriptional proteins, in driving the formation of multi-component transcriptional condensates [5,8–11]. In the crowded cellular milieu, these dynamic condensates are often spatially restricted to specific nuclear regions, influenced by the mechanical properties of surrounding chromatin, and exhibit atypical characteristics of a phase-separated system, such as aspherical morphology, sub-diffusive dynamics, and limited coarsening [12–15]. These observations have raised important questions regarding whether these assemblies are formed by canonical liquid–liquid phase separation and what are the factors that dictate the size, shape, and localization of these condensates [16–18].

In addition to the IDRs, transcription factors possess a DNA binding domain (DBD) that facilitates interactions with the chromatin. This interaction may position chromatin as a critical scaffold for the nucleation of DNA binding protein condensates in the cell nucleus. The characteristics of the resulting condensates can therefore be impacted by factors such as the number of available binding sites, DNA binding affinity, and local chromatin density. Studies on the KLF4 transcription factor and RNA polymerase II revealed that DNA promotes surface condensation at protein concentrations below their bulk threshold concentration for phase separation, known as saturation concentration [19,20]. Chromatin interactions have also been shown to be crucial for the nucleation of glucocorticoid receptor condensates and FET-family oncofusion condensation at specific genomic sites [21–23]. In other systems, contrasting observations were noted where loss of chromatin interactions was shown to enhance nuclear phase separation. For example, in the case of the NUP oncofusion, NUP98-HOXA9, the deletion of the DNA-binding domain of HOXA9 resulted in the formation of a fewer number of large spherical condensates whereas the wildtype protein formed numerous nanoscopic puncta in the nucleus [24]. Similarly, inhibiting the bromodomain-mediated BRD4 binding to acetylated chromatin using either a small molecule called JQ1 or truncating the bromodomains, leads to redistribution of BRD4 throughout the nucleoplasm and the formation of a few large foci [25]. This phenomenon was also observed with the protein ENL, where mutations in the YEATS domain, which lead to a loss of binding to acetylated histones enhance

its ability to form mesoscale condensates [26]. These observations collectively suggest a delicate interplay between protein–protein self-interactions driving homotypic phase separation versus protein–chromatin interactions, and the availability and density of binding sites on chromatin, that dictate the morphological diversity of nuclear condensates in cells. They also highlight the necessity of systematically mapping macromolecular condensation in live cells to capture their full range of behavior that may not be available from *in vitro* studies.

In this study, we employ live cell microscopy and molecular simulations to investigate how binding to chromatin influences the size and morphology of protein condensates. We focus on the intrinsically disordered prion-like domain (PLD) of the SS18 protein, a subunit of the mammalian SWI/SNF (mSWI/SNF) complex, and engineered *de novo* fusion proteins with tunable DNA binding ability. SS18^{PLD} readily undergoes phase separation with a saturation concentration of $\sim 2 \mu\text{M}$ *in vitro*, forming large ($\sim 2 \mu\text{m}$ or larger) spherical condensates in the regions of low chromatin density [27]. Fusion of SS18^{PLD} with the structured Forkhead (FKH) DNA-binding domain of the pioneer factor FOXA1 leads to condensates re-localizing on chromatin. Employing live cell imaging, we observe that the size and morphology of the SS18^{PLD}-FKH fusion protein condensates are significantly altered due to condensate–chromatin interactions. Utilizing selected mutants of FKH domain, we confirm that changes in condensate properties are driven by the FKH domain's chromatin binding affinity. We further employ sequence-resolved coarse-grained simulations and uncover a delicate interplay between PLD–PLD and FKH–DNA interactions that governs nuclear condensate morphology. In the absence of FKH domain, homotypic PLD interactions dominate over protein–DNA interactions, leading to the formation of spherical droplets. In contrast, FKH–DNA interactions dominate in SS18^{PLD}-FKH fusion, resulting in the formation of aspherical condensates on chromatin that display strongly sub-diffusive behavior. Simulations using mesoscale resolution coarse-grained models reveal that the condensates formed by SS18^{PLD}-FKH fusion extensively wet the chromatin chains.

Results and Discussion

Fusion to a DNA-binding domain alters the size and morphology of condensates formed by SS18^{PLD} in live cells

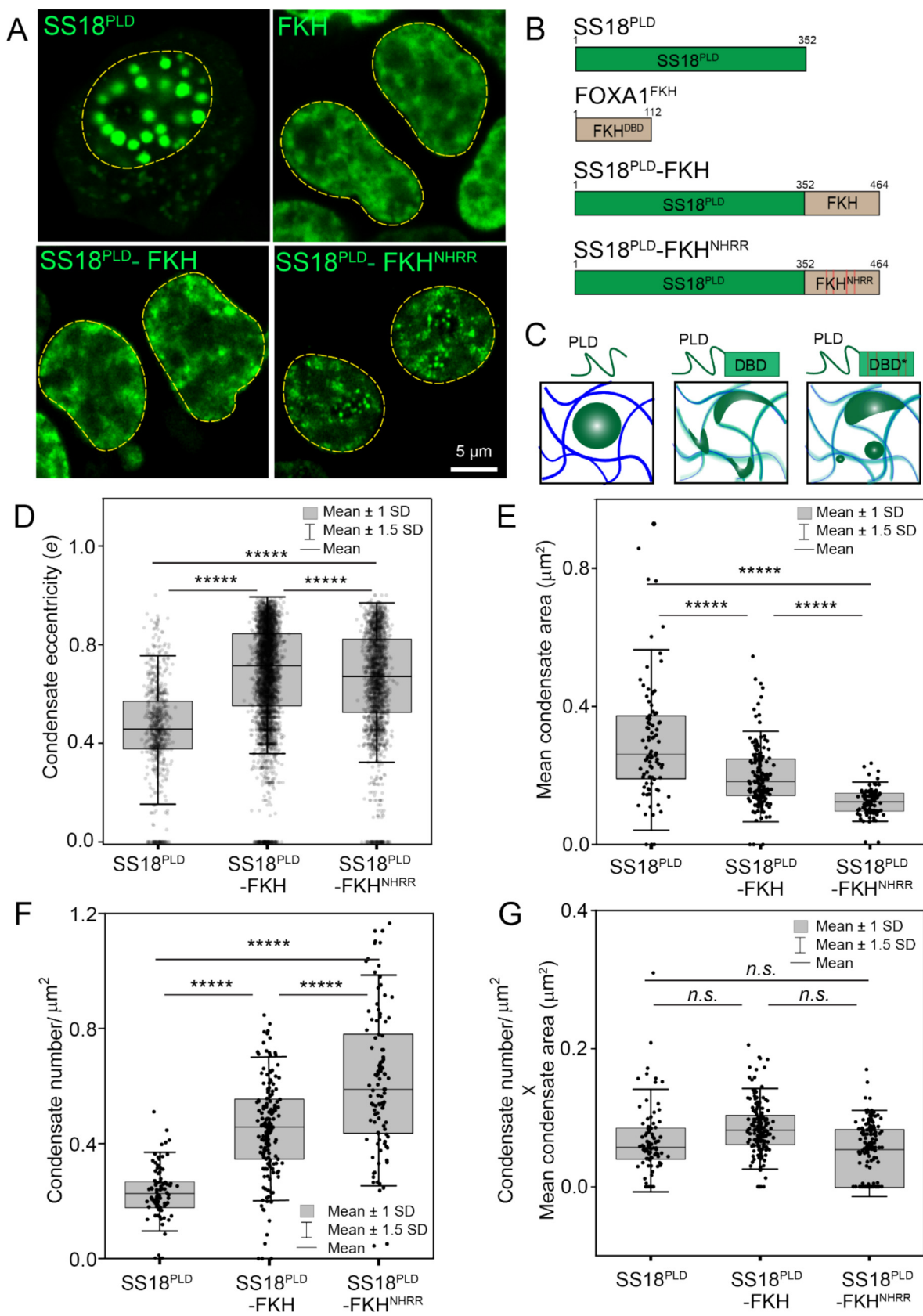
Transcription factors (TFs) typically exhibit diverse condensate morphologies within the nucleus ranging from nanoscale clusters to mesoscale condensates (Figure S1). To capture the range of such behavior, we transiently transfected HEK293T cells with eight GFP-labeled

protein constructs representing either full-length proteins or their selected domains (Figures 1A and S1). While expression of the GFP tag alone shows a very diffused distribution (Figure S1), we see that FUS-DDIT3, an oncogenic fusion TF containing a prion-like domain fused to a transcription factor from CCAAT/enhancer-binding protein (C/EBP) family [28,29], forms large, spherical condensates. In contrast, EWS-FLI1, another oncogenic fusion TF with a domain architecture similar to FUS-DDIT3, assembles into small cluster-like condensates resembling those formed by the RNA binding protein FUS (Figure S1A). Meanwhile, FOXA1, a pioneer TF, forms dynamic biomolecular condensates [13] that appear to be extensively wetting the chromatin with a few large foci. Despite the rich phase behavior of RNA and DNA-binding proteins that form condensates in the cell nucleus, the underlying molecular mechanism governing the size and morphology of these chromatin-bound condensates is unclear. Each transcription factor in Figure S1 possesses unique characteristics, such as differences in homotypic protein–protein interactions largely driven by the IDRs, as well as protein–chromatin interactions with varied binding affinity and binding site specificity.

These properties can collectively influence the formation of diffraction limited clusters (EWS-FLI1), large spherical condensates (FUS-DDIT3), or condensates wetting chromatin and appearing as protein clouds (FOXA1). To systematically address the role of chromatin binding on condensate morphology, we turned our attention to SS18, a subunit of the mammalian SWI/SNF complex. SS18 full-length (SS18^{FL}) protein forms condensates and its phase separation was shown to regulate pluripotent-somatic transition [30]. The phase separation of SS18^{FL} is driven by the intrinsically disordered prion-like domain (PLD: 67–418 aa). PLDs are IDRs that show low sequence complexity with a high enrichment of S, Q, Y, and G and are often depleted of charged residues [31–33]. PLDs typically are inert to nucleic acid binding but exhibit a strong propensity for homotypic condensation [34]. The condensates of SS18^{FL} are small and irregular both at low and high overexpression levels. The prion-like domains of SS18 (SS18^{PLD}), however, forms micron-sized condensates that have low asphericity and increase in size with protein expression levels. This behavior is also observed for other IDR systems that do not interact with the chromatin [35] (Figure S2). Time-lapse imaging of GFP-tagged SS18^{PLD} in HEK293T cells (Supplementary Movie 1, Figure S3A) revealed that the mean condensate area increases linearly with the protein concentration (Figure S3B), while condensate numbers initially grow steadily before plateauing at a later stage (Figure S3C). Throughout this process, condensates retain their sphericity (Figure S3D) and undergo coalescence (Figure S3E), which are expected behavior of phase

separated bodies with terminally viscous behavior. The transient expression of SS18^{PLD} in HEK293T cells can result in concentrations up to ~20 μ M (Figure S4) and consequently the non-physiological dominance of homotypic interactions. However, this simplified system provides an appropriate model for uncovering mechanistic insights on how condensate properties can be influenced by interactions with chromatin. To test the effect of chromatin binding on the condensation behavior of SS18^{PLD}, we created a chimeric construct by fusing the SS18^{PLD} with the Forkhead (FKH) DNA binding domain of FOXA1 referred to as SS18^{PLD}-FKH (Figure 1A and 1B). The FKH domain of FOXA1 can tightly bind to various chromatin sites either through specific or non-specific interactions [36,37]. We observe that upon being expressed in HEK293T cells, unlike SS18^{PLD}, SS18^{PLD}-FKH lacks the ability to form spherical condensates and instead forms irregular blob-like structures (Figure 1A and 1B). SS18^{PLD}-FKH condensate morphologies appear cloud-like, similar to that of FOXA1 assemblies (Figures 1A and S3A). We hypothesize that this can be a direct manifestation of the chromatin-wetting ability of SS18^{PLD}-FKH condensates due to the high affinity multi-site chromatin binding through the FKH domain (Figure 1C). If true, then chromatin wettability of condensates should sensitively depend on the chromatin binding strength at specific as well as non-specific DNA sites by the FKH domain [36,37]. Previous studies have established four point mutations in the FKH domain (FKH^{N216A, H220A, R262A, R265A}), known as the NHRR mutations, which disrupt the chromatin binding of FOXA1 [37]. Therefore, to test our hypothesis that reducing the affinity of FKH domain for chromatin may reduce the chromatin wettability of the SS18^{PLD}-FKH condensates, we transfected cells with a variant of SS18^{PLD}-FKH carrying the NHRR mutations (SS18^{PLD}-FKH^{NHRR}). Qualitatively, we observed that SS18^{PLD}-FKH^{NHRR} formed small spherical condensates in the cell nucleus and displayed substantially less wetting behavior on chromatin (Figure 1A). This data suggests that protein condensates can wet chromatin through favorable interactions encoded in the canonical DNA binding domains (DBDs) (Figure 1C).

To quantify the physical features of the condensates formed by SS18^{PLD}, SS18^{PLD}-FKH, and SS18^{PLD}-FKH^{NHRR}, we performed shape and size analysis using a custom-built image analysis toolbox that aided user bias-free foci segmentation (see *Methods*). Cells for quantitative analysis were selected where the mean nuclear intensity of the population for each variant is not statistically different from each other, thereby allowing us to compare condensate properties at similar overexpression levels (Figure S5). Eccentricity (e) measurements revealed that SS18^{PLD}-FKH ($e = 0.67 \pm 0.21$) condensates are predominantly aspherical, which is a signature of chromatin



wetting. In contrast, SS18^{PLD} ($e = 0.4 \pm 0.2$) forms dominantly spherical condensates (Figure 1D). We next quantified the size and number of condensates for each protein. The plots of mean condensate area and condensate number per unit area of nucleus (Figure 1E and 1F) reveal that SS18^{PLD} condensates were larger in size ($0.3 \pm 0.17 \mu\text{m}^2$) and fewer (0.23 ± 0.09 condensates/ μm^2) in number. In contrast, SS18^{PLD}-FKH condensates are smaller in size ($0.19 \pm 0.08 \mu\text{m}^2$) and more in number (0.45 ± 0.16 condensates/ μm^2). This is likely due to chromatin interactions through the FKH domain negatively regulating the coarsening of SS18^{PLD}-FKH condensates while the abundance of binding sites disperses the protein molecules. Interestingly, SS18^{PLD}-FKH^{NHRR} forms smaller ($0.12 \pm 0.03 \mu\text{m}^2$) condensates that are higher (0.6 ± 0.2 condensates/ μm^2) in number with a lower asphericity ($e = 0.63 \pm 0.21$) as compared to SS18^{PLD}-FKH. These observations can be attributed to the residual DNA binding affinity of NHRR mutant contributed by SW residues in the wing 1 domain of FKH [36,37], which may allow the SS18^{PLD}-FKH^{NHRR} condensates to be pinned at chromatin sites while simultaneously restricting their ability to undergo coarsening, resulting in higher number of smaller condensates. Additionally, the reduced DNA binding affinity leads to weakened chromatin wetting behavior resulting in condensates with lower eccentricity as compared to SS18^{PLD}-FKH condensates. However, we cannot rule out other possible mechanisms such as RNA binding [38] and/or microphase separation [39] that may underlie the observed differences in the size and morphology of the nuclear condensates formed by SS18^{PLD}-FKH and SS18^{PLD}-FKH^{NHRR}. Next, we plotted the product of the mean

condensate area and the condensate number per unit area which revealed no significant differences between the means of the three constructs (Figure 1G). This observation suggests that condensation propensity of SS18^{PLD}, SS18^{PLD}-FKH, and SS18^{PLD}-FKH^{NHRR} is likely similar, but condensate morphologies and coarsening dynamics are significantly influenced by the presence and characteristics of DNA/chromatin-interaction domains.

The CMV promoter used in our constructs is a strong promoter in HEK293T cells [40] which leads to greater than physiological protein concentrations (0.5–20 μM , Figure S4). Although we have not examined the phase behavior of these proteins at endogenous levels, we sought to investigate how varying expression levels influence condensate morphology in our overexpression-based cellular assay. We analyzed cells expressing either SS18^{PLD} or SS18^{PLD}-FKH with roughly threefold difference in expression levels. Utilizing fluorescence intensity as a proxy for concentration, we categorized cells as low (0.045–0.137 a.u.) or high (0.16–0.49 a.u.) expression. While condensate counts remained relatively unchanged for both SS18^{PLD} and SS18^{PLD}-FKH between either category (Figure S6A), the mean condensate area increased more significantly for SS18^{PLD} (2.5-fold) compared to SS18^{PLD}-FKH (1.9-fold) (Figure S6B). This suggests that the coarsening of SS18^{PLD}-FKH condensates is kinetically hindered compared to the SS18^{PLD} condensates, likely due to the spatial restrictions imposed by the FKH-chromatin interactions. This interpretation is further supported by trends in mean condensate eccentricity as a function of protein expression level: SS18^{PLD} condensates become more spherical, while morphologies of SS18^{PLD}-FKH become more irregular due to chromatin wetting effects (Figure S6C).



Figure 1. Chromatin binding impacts the size and morphology of protein condensates in cell nucleus. A. Fluorescence microscopy images of HEK293T cells transiently expressing GFP-tagged SS18^{PLD}, FKH, SS18^{PLD}-FKH, and SS18^{PLD}-FKH^{NHRR}, as indicated. The yellow dashed line represents the nuclear periphery. **B.** Domain architecture of SS18^{PLD}, FOXA1^{FKH}, SS18^{PLD}-FKH and SS18^{PLD}-FKH^{NHRR}. (PLD: Prion-like domain, FKH: Forkhead DNA binding domain). **C.** Schematic representing the chromatin binding interactions of SS18^{PLD}, SS18^{PLD}-FKH and SS18^{PLD}-FKH^{NHRR} (mutations disrupting sequence-specific and non-specific interactions in the FKH domain are highlighted in red lines). Chromatin is depicted in blue color. **D.** Box plot representing eccentricity (e) value of SS18^{PLD} condensates ($n = 705$ from 87 cells), SS18^{PLD}-FKH condensates ($n = 4098$ from 162 cells), SS18^{PLD}-FKH^{NHRR} condensates ($n = 2012$ from 103 cells). n represents the number of condensates analyzed. Significance was calculated by a student's two-tailed t -test. "*****" indicates a p -value less than 0.00001. Condensates were quantified by selecting cells within a nuclear mean intensity range from 0.02 to 0.16. **E.** Box plot representing mean condensate area per cell for SS18^{PLD} ($n = 87$), SS18^{PLD}-FKH ($n = 162$), SS18^{PLD}-FKH^{NHRR} ($n = 103$); n represents the number of cells analyzed. Significance was calculated by a student's two-tailed t -test. "*****" indicates a p -value less than 0.00001. **F.** Box plot representing condensate number per micron area of a cell for SS18^{PLD} ($n = 87$), SS18^{PLD}-FKH ($n = 161$), SS18^{PLD}-FKH^{NHRR} ($n = 103$); n represents number of cells analyzed. Significance was calculated by a student's two-tailed t -test. "*****" indicates a p -value less than 0.00001. **G.** Box plot representing the product of condensate number per unit nuclear area and the mean condensate area per nucleus for SS18^{PLD} ($n = 87$), SS18^{PLD}-FKH ($n = 162$), SS18^{PLD}-FKH^{NHRR} ($n = 103$); n represents number of cells analyzed. Significance was calculated by a student's two-tailed t -test. 'n.s.' indicates 'not significant' with a p -value greater than 0.05.

SS18^{PLD}-FKH condensates binding to chromatin leads to spatially restricted sub-diffusive behavior

To directly test whether the FKH-fusion constructs colocalize with chromatin, we co-expressed GFP-tagged versions SS18^{PLD}, FKH, SS18^{PLD}-FKH, and SS18^{PLD}-FKH^{NHRR}, or GFP alone, with mGarnet-tagged core histone protein H2B (Figure 2A). Utilizing a previously published protocol that probe chromatin binding of proteins in live cells via Pearson correlation coefficient (PCC) [41,42], we quantified the colocalization of the protein of interest with H2B (chromatin marker) within the nucleus of each cell (Figure 2B). We find that the SS18^{PLD} condensates primarily localize to regions devoid of H2B, resulting in a low PCC (0.05 ± 0.24). This is consistent with a previous report suggesting condensates formed by FUS prion-like domain tend to form and grow in low density chromatin regions [35]. However, fusion to the FKH domain, SS18^{PLD}-FKH (0.62 ± 0.09) significantly increased colocalization with H2B, with PCC values similar to the FKH domain alone (0.62 ± 0.09) (Figure 2B). In contrast, SS18^{PLD}-FKH^{NHRR} (PCC = 0.16 ± 0.26) showed diminished colocalization to the chromatin. The colocalization of these constructs were compared to GFP (-0.108 ± 0.2) which does not interact with chromatin (Figure 2B).

We reasoned that if chromatin binding regulates condensates wetting the chromatin, there should be a correlation between the chromatin binding strength and the condensate morphology, with aspherical morphologies being more pronounced for proteins with higher affinity for chromatin and vice versa. To test this idea, we next employed fluorescence recovery after photobleaching (FRAP) in live cells. Previous studies have established FRAP as a tool to measure the relative strengths of chromatin binding by transcription factors. Using quantitative FRAP measurements, these studies showed that TFs with stronger chromatin binding exhibit slower nuclear mobility and longer dwell times on chromatin [37,43]. Following a similar approach, we measured the half-time for maximum fluorescence recovery ($\tau_{1/2}$) and the fraction of immobile molecules of SS18^{PLD} alone and the fusion proteins (Figures 2C and S7A). For these measurements, the bleach area was kept constant for all samples to allow quantitative comparison (see Methods). We observed that a rapid fluorescence recovery for SS18^{PLD} condensates ($\tau_{1/2} = 2.92 \pm 1.004$), which is largely excluded from the chromatin. In contrast, SS18^{PLD}-FKH exhibited substantially slower recovery ($\tau_{1/2} = 7.88 \pm 4.39$), consistent with its ability to interact with chromatin. SS18^{PLD}-FKH^{NHRR} ($\tau_{1/2} = 3.12 \pm 0.67$) with weakened chromatin-binding ability, displayed a fluorescence recovery dynamics similar to SS18^{PLD}. The FKH domain alone showed rapid recovery ($\tau_{1/2} = 0.57 \pm 0.27$), suggesting that the condensate formation by

SS18^{PLD}-FKH slows down macromolecular diffusion in the nucleus and increases their dwell times on specific genomic loci [44]. We also estimated the immobile fractions within the time frame of our analysis (30 s). We observe that the SS18^{PLD}-FKH had $\sim 40\%$ of its molecules bound to chromatin whereas the other constructs showed less than 10% chromatin-bound population (Figure S7B). These findings support the conclusion that fusion of the FKH domain to SS18^{PLD} directs it to chromatin leading to its restricted diffusion. In contrast, mutations within the FKH domain impair its DNA binding ability consequently increased diffusion (Figure 2D), thereby regulating the condensation landscape.

Molecular simulations reveal mechanistic insights into the effect of chromatin binding on protein condensation

To gain deeper insight into how chromatin binding impacts formation and coarsening of SS18^{PLD}-FKH condensates with DNA, we employed sequence-resolved coarse-grained simulations. Using the hydrophobicity scale (HPS) model [45–47] with an elastic network applied to the Forkhead domain and stacking interactions applied to the DNA, we first simulated a single-DNA system mixed with either SS18^{PLD}, SS18^{PLD}-FKH, SS18^{PLD}-FKH^{NH} (FKH mutation that results in the loss of sequence-specific DNA interactions but preserves sequence non-specific interactions) or SS18^{PLD}-FKH^{NHRR} (FKH mutation that results in the loss of both sequence-specific and non-specific DNA interactions) at moderate concentrations ($\rho = 1.3$ mM) (Figure 3A). In this single-DNA setup, we observed that the SS18^{PLD} promotes the formation of a PLD-rich dense phase fully separated from the DNA. In contrast, simulations with SS18^{PLD}-FKH proteins resulted in a mixed dense phase where DNA is surrounded by FKH domains (Figure 3A). By introducing various mutants (Figure 3A–B) and modulating the salt concentration of the solution (Figures 3C and S8), we effectively tuned the electrostatic component of DNA-protein interactions. For generating DNA affinity comparable to experimental observations for intracellular condensates formed by various mutants, a salt concentration of $\rho_{\text{NaCl}} = 0.25$ M is selected for the sequence-resolved coarse-grained simulations. At lower salt concentrations typically chosen for *in vitro* settings (e.g., $\rho_{\text{NaCl}} = 0.17$ M), DNA binding is expected to be stronger, which could result in reduced dissolution of protein-DNA condensates in Figure 3D, particularly for the NHRR mutant leading to higher colocalization and lower eccentricity. Simulation snapshots (Figure 3A) and the associated radial density plots (Figure 3B) illustrate tighter protein condensation with DNA for wild type compared to mutants and SS18^{PLD}.

Next, to explore how the interplay between DNA binding affinity and PLD self-interactions

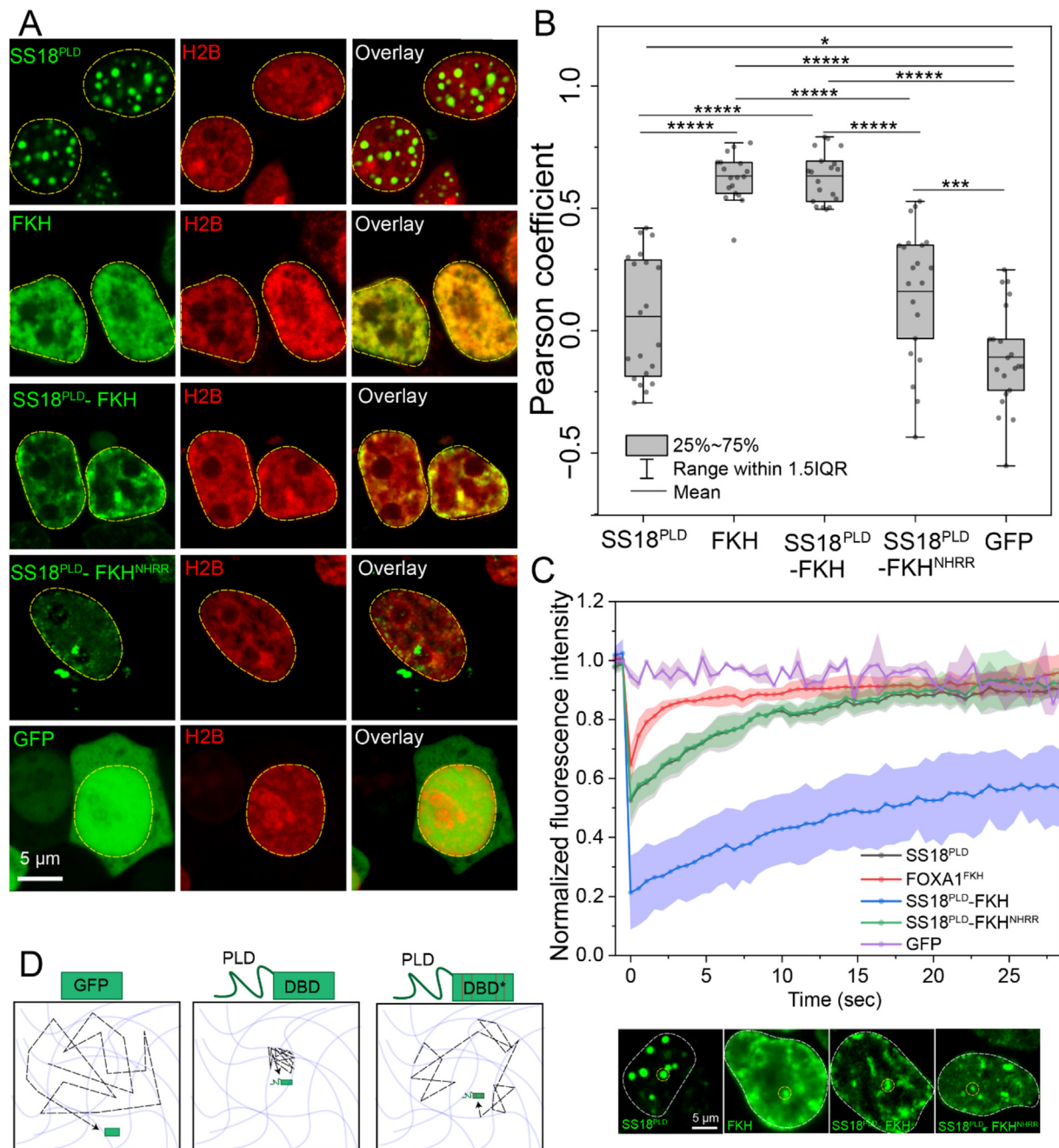


Figure 2. Fusion of the DNA-binding Forkhead domain (FKH) retargets SS18^{PLD} condensates to the chromatin and restricts their nuclear mobility. **A.** Fluorescence microscopy images of HEK293T cells transiently expressing either GFP-tagged SS18^{PLD}, FKH, SS18^{PLD}-FKH, SS18^{PLD}-FKH^{NHRR}, and GFP alone in conjunction with mGarnet-tagged H2B. The yellow dashed line represents the nuclear periphery. **B.** Box plot displaying the Pearson correlation coefficient (PCC) value of each cell co-expressing either SS18^{PLD}, FKH, or SS18^{PLD}-FKH, or SS18^{PLD}-FKH^{NHRR}, or GFP with H2B-mGarnet ($n = 12$ cells). Significance was calculated by a student's two-tailed t -test. “*” indicates a p -value < 0.05 . “***” indicates a p -value < 0.001 . “*****” indicates a p -value < 0.00001 . **C.** FRAP curves for SS18^{PLD}, FKH, SS18^{PLD}-FKH, SS18^{PLD}-FKH^{NHRR}, and GFP. The average intensity and standard deviation of the intensity profiles are shown over time ($n = 5$ cells). Representative cell images for each construct show the condensates that were bleached (highlighted by yellow circles). **D.** Schematic illustrating molecules with varying diffusion behavior: unrestricted diffusion (GFP), diffusion restricted by chromatin interactions (SS18^{PLD}-FKH), and intermediate diffusion due to weakened chromatin interactions (SS18^{PLD}-FKH^{NHRR}). Chromatin is shown in blue.

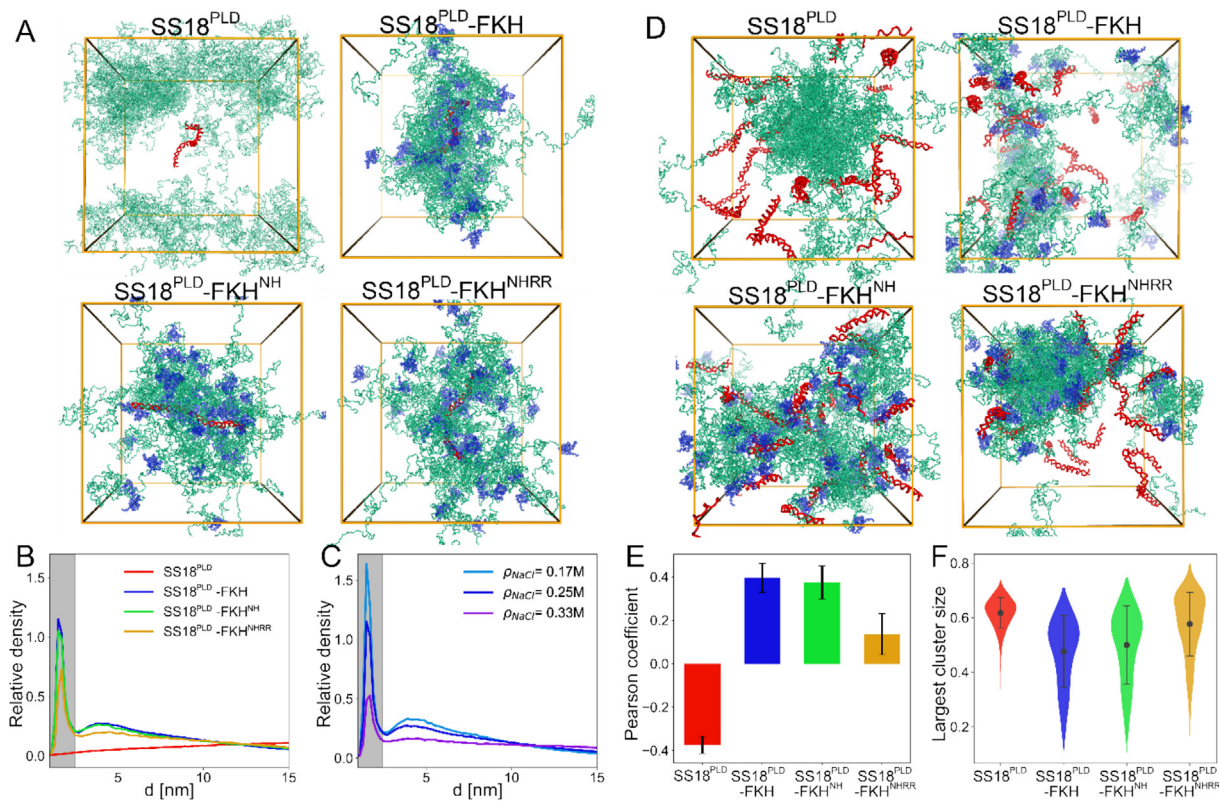


Figure 3. Coarse-grained (CG) simulations of single and multiple DNAs with proteins, highlighting changes in local affinity, colocalization, and clustering behavior. **A.** Snapshots of a single DNA molecule interacting with various protein constructs (SS18^{PLD}, SS18^{PLD}-FKH, SS18^{PLD}-FKH^{NH} and SS18^{PLD}-FKH^{NHRR}). The DNA is depicted as a red double helix. The PLD regions of proteins are represented as green lines and binding domains are represented as blue spheres. **B.** Radial distribution of proteins residues relative to the DNA backbone for 0.25 M salt concentration. The shaded grey area indicates the binding region around DNA defined by the first minima of density profile, and the peaks (approximately $d = 1.5$ nm) represent binding affinity strength. **C.** Radial distribution of protein residues relative to the DNA backbone for varying salt concentrations. **D.** Snapshots of simulations showing multiple DNA segments interacting with proteins. **E.** Representative plot of Pearson correlation coefficients, illustrating the degree of colocalization of DNA and proteins. Their averages represented by bars and standard deviations are represented by error bars. **F.** Representative violin plot of the largest cluster size, showing the distribution of the normalized largest cluster size. Their averages and standard deviations are represented by black dots and error bars.

influences condensate morphologies, we conducted multi-DNA simulations at higher protein concentrations ($\rho = 1.8$ mM). Introducing FKH domain mutations that weaken DNA binding, significantly impacted colocalization and cluster morphology, as shown in Figure 3D. In the absence of DNA binding domains (SS18^{PLD}), DNA and proteins remained well-separated, forming spherical SS18^{PLD} condensates. However, introducing FKH domains and increasing DNA affinity led to greater degree of colocalization between the DNA and protein chains, resulting in increasingly irregular condensate shapes. DNA-protein colocalization was quantified using Pearson correlation coefficient (Figure 3E). At the same bulk density of simulations, clusters with the lower eccentricity exhibit a stronger tendency to form condensates, corresponding to larger cluster sizes. The cluster size variability was analyzed by

examining the distribution of the normalized largest cluster size (Figure 3F). The trend aligns with the eccentricity observed in Figure 1D. We also performed contact map analysis of condensates composed of DNA and chimeric proteins containing PLD and DBD domains (Figure S9). This analysis provides direct insights into the relative interaction strengths of different domains within the condensates. As expected, SS18^{PLD} exhibits stronger protein-protein interactions compared to protein-DNA interactions. Comparing SS18^{PLD}-FKH, SS18^{PLD}-FKH^{NH} and SS18^{PLD}-FKH^{NHRR}, which include a DBD with variable DNA binding affinity, our analysis reveals the contacts between DBD and DNA are significantly stronger than other regions for SS18^{PLD}-FKH. Mutations in the DBD (e.g. in FKH^{NHRR}) reduce these contacts compared to the original FKH construct.

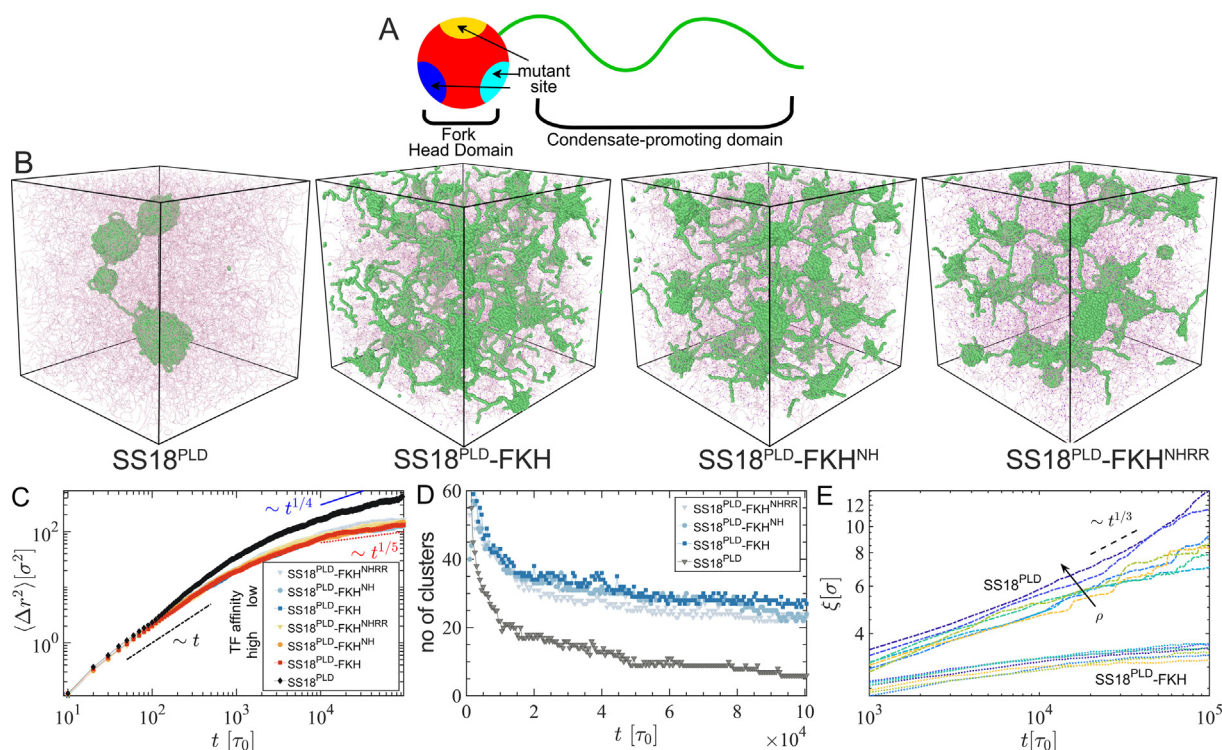


Figure 4. Mesoscale resolution coarse-grained simulations of transcription factors in a dense mesh of chromatin network. **A.** The mesoscale model of the Forkhead (FKH) domain is represented as a patchy rigid particle with three chromatin binding sites. SS18^{PLD} is represented as a disordered flexible chain. **B.** Morphologies resulting from simulations of chromatin network mixed with protein constructs as indicated in the figure. The chromatin network is colored transparent purple to aid in visualizing morphologies of transcription factor condensates. **C.** The mean squared displacement, $\langle \Delta r^2 \rangle$, as a function of time, contrasting diffusive scaling at long times for transcription factors with high (red shaded) and low (blue shaded) DNA-binding affinities. **D.** Cluster number as a function of time during phase separation. Starting from a mixed initial state, the number of clusters decreases over time as domains coarsen. SS18^{PLD} equilibrates into fewer, larger spherical droplets compared to the constructs with FKH domain that form spatially arrested microdroplet state. **E.** Time evolution of droplet growth characterized by the correlation length ξ . For SS18^{PLD}, ξ , increases with time as droplets grow and eventually coalesce into a single large domain, after which it plateaus. In contrast, the presence of FKH limits coarsening, leading to early saturation at smaller droplet sizes. Increasing the number density of protein, ρ , increases domain size ξ , with the constant rate of droplet growth. The different colored lines represent different protein number densities, where yellow indicates lower density and darker colors correspond to higher densities.

To evaluate the impact of condensate morphologies on the diffusive dynamics of TFs within chromatin-like viscoelastic microenvironments, we conducted simulations using a mesoscale-resolution coarse-grained model. In this framework, SS18^{PLD} is modeled as an unstructured coarse-grained polymer chain with a tendency to undergo phase separation due to weak self-attractive interactions, while the FKH domain is represented as a patchy particle containing three specific binding sites with high affinity for chromatin (Figure 4A). Chromatin itself is modeled as a dense polymer mesh with weakly attractive interactions restricted to the DNA-binding patchy sites. There are two primary scenarios that arise: (i) in the absence of the FKH domain, SS18^{PLD} phase forms a macroscopic cavity within the chromatin network (Figure 4B).

This cavity does not wet the chromatin, leading to an elastic deformation energy penalty associated with network distortion [48,49]. (ii) In contrast, when the FKH DNA-binding domain is present, FKH–chromatin interactions arrest phase separation dynamics of SS18^{PLD}, resulting in the formation of numerous small, spatially confined microdroplets within the chromatin network. SS18^{PLD}-FKH droplets appear to wet the chromatin network due to favorable protein–DNA interactions at the droplet–chromatin interface. This morphological difference is illustrated in the simulation snapshots shown in Figure S10, where protein condensates either exclude or wet the chromatin surface, depending on the strength of protein–chromatin interactions. Consistent with experimental results (Figure 1), our simulations show that the FKH domain in the SS18^{PLD}-FKH fusion protein induces extensive

binding to the chromatin network (Figure 4B). This chromatin-FKH interactions leads to the formation of protein cloud-like morphologies in contrast to the spherical droplets observed in the case of SS18^{PLD} alone. The incorporation of FKH binding sites shifts the equilibrium toward a percolated network-like state, as opposed to discrete spherical droplet morphologies (Figure 4B). This transition is quantitatively captured by the spatial distribution of protein cluster radii (R_g) and their connectivity within the chromatin network (see *Methods*). Notably, we observe a change in the diffusive dynamics in the phase separation process, as reflected in mean squared displacement (MSD) profiles of center of mass coordinates of transcription factors (Figure 4C). The extracted power-law exponent (α) of the MSD $\langle \Delta r^2 \rangle \sim t^\alpha$ reveals normal diffusive dynamics at the early time of the phase separation, where $\alpha = 1$. However, at later times, in equilibrium, SS18^{PLD} diffusion further transitions into an arrested dynamical regime ($\alpha \approx 0.25$), indicating that PLD condensation inside the chromatin cavity progressively restricts its mobility. On the other hand, high DNA-binding affinity, as is the case for SS18^{PLD}-FKH fusion protein, leads to the formation of smaller clusters with increasingly arrested dynamics, which manifests in a shift towards more extreme subdiffusive behavior ($\alpha \approx 0.2$). Overall, the number of condensate domains in equilibrium for designed TF constructs with variable DNA binding affinity corroborates the size and morphology of condensates observed in our experiments (Figure 1). Starting from a homogeneous mixed state, the system undergoes phase separation, during which the number of clusters progressively decreases as individual domains grow and coarsen over time (Figure 4D). For SS18^{PLD}, this coarsening process leads to the formation of fewer but larger and more spherical condensates, indicating efficient droplet growth. In contrast, the presence of the FKH domain results in a more fragmented condensate morphology, with numerous smaller clusters that persist over time, suggesting that FKH-chromatin interactions inhibit coarsening and maintain a more dispersed, microdroplet-like state. To quantify the kinetics of phase separation and coarsening, we compute the time-dependent correlation length, $\xi(t)$, from the spatial two-point correlation function $C(r, t) = \langle \phi(\mathbf{x}, t) \phi(\mathbf{x} + \mathbf{r}, t) \rangle - \langle \phi(\mathbf{x}, t) \rangle \langle \phi(\mathbf{x} + \mathbf{r}, t) \rangle$, which quantifies how spatial fluctuations in coarse-grained order parameter field $\phi(r, t)$ (the PLD phase) at time t are correlated across distance r . We extract the correlation length $\xi(t)$ as the exponential decay length of $C(r, t)$, representing the typical size of phase-separated domains at time t . The correlation length grows with time following a power law: $\xi(t) \sim t^\beta$, where the exponent β depends on the mechanism of coarsening [50]. For the SS18^{PLD} system (Figure 4E), we observe typical diffusion-limited coarsening, $\beta \approx 1/3$, corresponding to droplet growth and eventual coalescence into

a macroscopic domain. Within our simulation time-scales, for smaller densities of SS18^{PLD} systems, $\xi(t)$ reaches plateaus, indicating an equilibrium state. In contrast, the SS18^{PLD}-FKH system exhibits early saturation of $\xi(t)$, reflecting suppressed coarsening due to strong, spatially restricted interactions between the FKH domain and the DNA network. These interactions inhibit droplet coalescence and lead to the formation of multiple small, spatially pinned condensates. We also compute the surface area (S) of condensates (using alphaShape algorithm) at equilibrium [51], which increases linearly with protein number density for both SS18^{PLD} and SS18^{PLD}-FKH. Consistent with suppressed coarsening dynamics, we observe that SS18^{PLD} forms fewer, larger droplets, while SS18^{PLD}-FKH yields many smaller clusters due to high affinity Forkhead-chromatin binding, resulting in a thermodynamically stable, spatially arrested microdroplet state (Figure S11).

To further probe the role of FKH-chromatin interactions, we introduced a mutation in the SS18^{PLD}-FKH construct (SS18^{PLD}-FKH^{NH}), which disrupts the sequence-specific interaction between the Forkhead domain and chromatin. As a result, the restriction imposed by chromatin binding is relieved, leading to the formation of larger condensates compared to SS18^{PLD}-FKH (Figure S12). Equilibrium snapshots from our simulations (inset) highlight this transition, where the number of small cluster nodes decreases while R_g increases upon mutation, indicating a shift from a dense, network-like structure toward a more dispersed morphology (Figure S12). Further mutations (SS18^{PLD}-FKH^{NH} and SS18^{PLD}-FKH^{NHRR}) lead to a continued increase in R_g driving the system from a chromatin-associated network toward droplet-like condensates, which is consistent with our experimental results (Figure 1). These results indicate that the balance between self-interaction among PLDs and heterotypic chromatin-binding interactions play a key role in determining condensate morphology. When self-attractive interactions dominate, condensates take nearly spherical shapes in the absence of chromatin binding. Conversely, when strong FKH-DNA interactions are introduced, condensates adopt network-like morphologies with extensive chromatin-TF association and higher cluster size variability. At higher protein concentrations, increasing DNA-binding affinity significantly enhances DNA-protein colocalization, leading to a shift in condensate morphology toward cloud-like, network-enriched structures. These findings from CG model provide a quantitative framework for our experimental observations (Figures 1 and 2) and underscore the critical role of interaction strengths and chromatin accessibility in shaping condensate phase behavior. This could have functional implications in transcriptional regulation by

modulating the spatial organization and mobility of transcription factors within chromatin environments.

Conclusion

Recent studies have highlighted the role of phase separation of coactivators, transcription factors and chromatin remodelers in the context of transcription regulation and disease processes [2,8,9,28,52–54]. In many instances, however, transcription factors within cells seldom form the characteristic large spherical droplet morphologies expected based on the *in vitro* assays. Instead, their morphologies resemble microphase-separated domains, suggesting that the nuclear microenvironment and transcriptional activity play a critical role in condensate formation and regulation [12,13,55,56].

In this study, we elucidated the effect of chromatin binding on condensate morphology. We show that despite the presence of a strong driver of intracellular phase separation via a prion-like domain (PLD), engineered PLD-containing fusion proteins with a strong DNA-binding domain form aspherical and spatially restricted condensates that appear as “protein clouds”. Through experiments and simulations, we confirm that this behavior is correlated with their ability to wet the chromatin network through the DNA-binding domain (Figure 1A). This finding raises important questions about the role of chromatin in shaping the phase behavior of DNA- and chromatin-binding proteins *in vivo*.

Sequence-specific interactions have been shown to drive the formation of transcription factor condensates at specific chromatin-binding sites [19,23,25,57]. In our system, SS18^{PLD} forms large spherical condensates while the SS18^{PLD}-FKH fusion protein forms many small bright foci in the live cell nucleus that remain tethered to chromatin. These loci likely represent condensate nucleation sites due to sequence-specific and non-specific chromatin interactions. SS18^{PLD}-FKH condensates display low diffusion rates in FRAP assays compared to SS18^{PLD} or the FKH domain alone (Figures 2C and S7) and high colocalization with histone H2B, a chromatin marker (Figure 2A–B). These observations support the view that chromatin tethering inhibits bulk phase separation of the SS18^{PLD}-FKH protein. Moreover, the FKH domain can also non-specifically target the SS18^{PLD} to chromatin, resulting in surface condensation of the fusion protein that mirrors observations made for KLF4 *in vitro* [19] and RNA polymerase II in cells [20]. Notably, the oncofusion transcription factor FUS-DDIT3 formed large, spherical condensates in HEK293T cells upon overexpression, despite containing a DNA-binding domain (DBD), in contrast to EWS-FLI1, a TF with similar domain architecture, which formed small, irregular puncta under similar conditions (Figure S1). However, in its native context within an MLS cell line, FUS-

DDIT3 forms small, irregular puncta [29]. Investigation of FUS-DDIT3 condensates across low and high expressions showed that similar to SS18^{PLD}, FUS-DDIT3 condensates increased significantly in size with higher expression while retaining their spherical morphology (Figure S13). This expression-dependent growth, coupled with low asphericity, suggests that FUS-DDIT3 condensates formed via protein over-expression may not engage with chromatin in HEK293T cells, likely due to inaccessibility of the specific chromatin-binding sites or cofactors required for native interactions. These observations collectively suggest microdroplet-like morphologies of chromatin-associated factors in cells are likely surface condensation or adsorption on the chromatin surface that limits growth but can retain liquid-like characteristics.

Overall, using experiments and simulations, this work establishes a broadly applicable physical framework for understanding how the domain architecture of TFs and their chromatin interactions dictate condensate morphology, dynamics, and spatial regulation in the nuclear environment. While our overexpression-based approach enabled detailed characterization of these relationships, the resulting supraphysiological protein concentrations represent a key limitation. Future studies can explore the dynamics, morphology, and material properties of nuclear condensates under endogenous protein concentrations in native cell lines, and the role of non-equilibrium processes including active transcription [55,56] in regulating the condensation landscape of multivalent DNA-binding proteins.

Methods

Cell culture

The HEK293T cells were cultured at 37 °C and 5% CO₂ in Dulbecco's Modified Eagle's Medium supplemented with 10% fetal bovine serum. To initiate transfection, 20,000 cells were seeded in 8 well-chambered slides. After 24 h, Lipofectamine 2000 reagent was used to transfect the cells with 0.5 µg plasmid, according to the manufacturer's protocol. The cells were imaged after 16–18 h of transfection. Colocalization experiments were carried out with 0.5 µg plasmid for each construct. Table S1 provides a complete list of proteins used in this study and their amino acid sequences.

Fluorescence imaging

Cells were incubated in Fluorobrite media supplemented with 10%(v/v) FBS and Hoechst 33342 dye to stain the nucleus. Imaging was carried out using either the Andor Dragonfly Spinning Disk microscope (63× objective), Leica Stellaris (63× objective), or Q2 (ISS Inc., 63×

objective) laser scanning confocal microscopes, with the cells being maintained at 37 °C on a temperature and CO₂-controlled stage. The plasmid constructs with mEGFP (Addgene: #54759) and H2B-mGarnet (Addgene Plasmid #104311) were imaged using 488 nm laser and 594 nm laser respectively, while Hoechst imaging was done using the 405 nm laser. Time-lapse imaging for SS18^{PLD} condensates were performed 2 h after transfection using Q2 (ISS Inc., 63× objective) laser scanning confocal microscope for a duration of 6 h.

To determine the nuclear protein concentrations in overexpressed system, we performed mEGFP (GFP henceforth) titration assay. The mean fluorescence intensity (a.u.) (y-axis) was measured using purified GFP protein across different concentration range (0.75–27 μM) (x-axis) under the same image acquisition settings used for cells. Linear regression analysis revealed the following relationship where x represent GFP concentration (in μM) and y represents mean fluorescence intensity: $y = 0.0013x + 0.0105$.

Image analysis

For the quantification of biomolecular condensates in our study, we employed a custom in-house image analysis platform developed in Python. Raw fluorescence images were first pre-processed to enhance signal quality. Pre-processing included noise suppression via wavelet decomposition and Gaussian filtering, as well as feature enhancement through Laplacian of Gaussian filtering and morphological operations, including top-hat filtering. These steps served to minimize noise and artifacts while enhancing the fidelity of objects present in the images. Background removal was then performed before applying a multi-tiered segmentation strategy and a multi-faceted object filter to eliminate false positives. An integrated human-in-the-loop verification step further refined segmentation by allowing manual exclusion of spurious signals and artifacts. For colocalization analysis, image-analysis platform performed pixel-based correlation analyses between multiple fluorescence channels. Specifically, Pearson's correlation coefficient was calculated for each channel pair to quantify the degree of spatial overlap and intensity correlation.

For analysis of time-lapse images of SS18^{PLD}, the condensates in a frame were segmented and analyzed using CellProfiler [58]. Once segmented, each condensate was characterized by a range of metrics, including area, eccentricity, and integrated fluorescence intensity. These measurements provide detailed insights into condensate size, shape, and protein concentration, forming the basis for downstream statistical analyses.

FRAP (Fluorescence recovery after photobleaching) measurements

FRAP experiments were performed using Leica Stellaris laser scanning confocal microscope. A circular region of 0.3 μm radii was photobleached using three lasers set at 100% laser intensity for 3 s followed by a post-bleaching imaging scan for 30 s. The GFP intensity values from the bleached region of interest were then corrected for photofading by normalizing them to intensities from an unbleached reference condensate. The FRAP recovery curves were plotted using OriginPro2024 and fitted using the equation $I(t) = a(1 - \exp(-\frac{t}{\tau}))$ as previously reported [59].

Sequence resolved simulations of protein-DNA condensates

We have employed a one-bead-per-residue hydrophobicity scale (HPS) model for co-condensates of double-strand DNA and chimeric protein SS18^{PLD}-FKH, consisting of a prion-like domain (PLD) of SS18 and a forkhead (FKH) DNA binding domain of FOXA1. The CG beads in the DNA chain represent adenine-base (A) and thymine-base (T) nucleotides, and the beads in the protein represent amino acids. The non-bonded interactions between residues i, j are modeled via the Ashbaugh-Hatch potential, and salt-screened electrostatic interactions are modeled via the Debye-Huckel potential:

$$U_{ij}^{AH}(r) = \begin{cases} U^L(r) - \lambda_{ij}U^L(r_c) + \epsilon(1 - \lambda_{ij}), & r \leq 2^{1/6}\sigma_{ij} \\ \lambda_{ij}[U^L(r) - U^L(r_c)], & 2^{1/6}\sigma_{ij} < r \leq r_c \\ 0, & r > r_c, \end{cases}$$

$$U_{ij}^{DH}(r) = \frac{q_i q_j e^2}{4\pi\epsilon} \frac{\exp(-\kappa r)}{r}$$

where $\epsilon = 0.8368$ kJ/mol, $r_c = 4$ nm, and U^L is the Lennard-Jones potential $U^L(r) = 4\epsilon[(\sigma_{ij}/r)^{12} - (\sigma_{ij}/r)^6]$. $\sigma_{ij} = (\sigma_i + \sigma_j)/2$ and $\lambda_{ij} = (\lambda_i + \lambda_j)/2$ are arithmetic averages of monomer size and hydrophobicity value, respectively. The pair potential with $\lambda = 0$ consists of only the repulsive term equivalent to the Weeks-Chandler-Andersen functional form. We applied σ of amino acids from van der Waals volumes and λ of amino acids from the recently proposed CALVADOS 2 parameters [47]. σ and λ of nucleotides come from previous work [60,61]. In the Debye-Huckel potential, q represents the charge number of particles, and Debye length is defined as $\kappa^{-1} = \sqrt{\epsilon k_B T / 2e^2 I}$, where ϵ is the dielectric constant and I is the ionic strength. The Debye length is approximately 0.9 nm in our simulations. The temperature-dependent dielectric constant has the following empirical relation: $(T)/\epsilon_0 = 5321/T + 233.76 - 0.9297T + 1.41710 \times 10^{-3}T^2 - 8.29210 \times 10^{-7}T^3$. All CG beads are

connected via bonded interactions modeled by harmonic potentials $U^{bond}(r) = \frac{k}{2}(r - r_0)^2$ where $k = 8033$ kJ/mol/nm², $r_0 = 0.5$ nm for DNA and $r_0 = 0.38$ nm for peptide. There are additional base-pair interactions and stacking interactions between adjacent base-pair groups in DNA. The base pair interaction is represented by repulsive Morse potential $U^{bp}(r) = D \exp(-\alpha(r - l_0))$ with $D = 472$ kJ/mol, $\alpha = 2.876$ nm⁻¹ and $l_0 = 1.243$ nm. The stacking interaction is modeled using a Lennard-Jones potential, characterized by $\epsilon = 49.2$ kJ/mol, $\sigma = 0.2938$ nm, and a cut-off distance $r_c = 5.65$ nm. We used an elastic network model (ENM) with harmonic potential to restrain non-bonded pairs in the folded domains of protein using $U^{ENM}(r) = \frac{k_d}{2}(r - r_0)^2$. Here, the force constant $k_d = 700$ kJ/mol/nm². We only apply the ENM to residue pairs with an r_0 below 0.9 nm cutoff. The predetermined configuration and boundary of the folded domain are generated by the alphaFold2 colab.

We performed our coarse-grained (CG) simulations using OpenMM (v7.7). Simulations were conducted in a cubic box with a side length of 40 nm. For single-DNA simulations, we set the temperature $T = 360$ K, protein bead concentration $\rho = 0.6$ M, and ionic strength $I = 170 - 330$ mM (same as the NaCl salt concentration). The DNA molecule was 100 base pairs long, and approximately 60 proteins were included, with the exact number depending on their sequence lengths. For multi-DNA simulations, we used $T = 340$ K, $\rho = 0.8$ M, and $I = 250$ mM. Each DNA molecule was 40 base pairs long, with 24 DNA molecules included in the system. Around 80 proteins were added, with the exact number again dependent on sequence length. After energy minimization, all systems were simulated in the NVT ensemble with a timestep of 0.01 ps. Each simulation was run for 5×10^7 steps. We applied the DBSCAN clustering algorithm to group protein data points. Each point represents the center of approximately 50 CG residues, with slight variations depending on protein length. The minimum number of points required to define a core point was set to 7, and the neighborhood distance parameter (ϵ) was set to 3.5 nm.

Contact map analysis

We used MDAnalysis python package to compute contact maps for protein–protein and protein–DNA interactions [62]. For each frame of the trajectory, contacts were calculated using each protein residue as a reference. The analysis was averaged over 100 frames sampled every 200 ns after equilibration. A cutoff distance of $d = 1.5$ nm, determined from the radial distribution function (Figure 3B), was used to define contacts. The contact frequency relative to a protein chain (bottom panel

in Figure S9) was obtained by averaging each column of the contact map.

Mesoscale simulations of transcriptional condensates in chromatin network

We conducted mesoscale coarse-grained (CG) molecular dynamics (MD) simulations to explore the condensate droplets and chromatin network structures for chimeric protein SS18^{PLD}-FKH and SS18^{PLD}. First, we simulated conditions replicating the experiment for SS18^{PLD} to capture spherical condensate. Next, we employ a spherical head domain, modeled as an independent rigid body with three patches on its surface mimicking DNA binding sites interacting with chromatin sites. One end of the protein chain is attached to the head segment. The condensate-promoting proteins are represented by chains of 100 CG beads, which interact via an attractive LJ potential with an energy depth of ϵ . Additionally, we incorporate long chromatin chains 3000 CG beads with specific binding sites for the fork-head. Molecular dynamics simulations were performed using the LAMMPS package. The equations of motion were integrated within the NVT ensemble for chromatin and protein chains and NVT/rigid for the head domain, employing a Langevin thermostat with a time step of 0.005τ , where $\tau = \sqrt{m\sigma^2/\epsilon}$ represents the derived unit of time, and m denotes the mass of a bead and σ denotes the bead diameter and establishes the length scale for LJ simulations. The Langevin thermostat ran 10^7 steps and was configured with a friction factor of 0.1τ . DNA binding sites formed bonds with specific sites of the chromatin if the radius is within the cutoff distance $r_b < 1.3\sigma$. Initially, systems were homogeneously distributed at high temperatures. Non-bonded interactions were incorporated using the attractive Lennard-Jones (LJ) potential for condensate-promoting protein domains. For other non-bonded interactions, a truncated LJ (WCA) repulsive interaction was implemented. To mimic the introduction of mutant sites, we switched off the bond formation of specific sites, transitioning the system from a cloud network to completely spherical droplets. There are 20 chromatin CG chains, and 200 protein chains initially positioned randomly within a periodic cubic simulation box with edge lengths of $L_x = L_y = L_z = 100$. The chains of biomolecule beads are connected by anharmonic FENE potential $U_{FENE}(r) = -\frac{1}{2}kR_0^2 \ln(1 - r^2/R_0^2)$, where r represents the distance between adjacent beads, $k = 20\epsilon/\sigma^2$ is the spring constant, and $R_0 = 1.5$ is the maximum extensibility. Bond angles between consecutive beads were constrained by an angular harmonic potential, $U_\theta = K_\theta(\theta - \theta_0)^2$, where K_θ denotes the angular potential constant and $\theta_0 = 150^\circ$ is the equilibrium angle.

CRedit authorship contribution statement

Anushka Supakar: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Richoo B. Davis:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Subhadip Biswas:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation. **Sean Yang:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation. **Davit A. Potoyan:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Investigation, Funding acquisition. **Priya R. Banerjee:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

DATA AVAILABILITY

All data are included either in the main text or in the supplementary information.

DECLARATION OF COMPETING INTEREST

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: “P.R. B. is a member of the Biophysics Reviews (AIP Publishing) editorial board. This affiliation did not influence the work reported here. All other authors have no conflicts to report.”.

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Appendix A. Supplementary material

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† These authors contributed equally.

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